Foundation Models for Precision Classification in NSBI: toward a Higgs self-coupling (κ_λ) measurement in $HH \to b\bar{b}\tau\tau$

Jing Pan (working note — draft for discussion)

19 June 2026

Bottom line

For constraining κ_λ, the statistically-limited piece is *kinematic shape* discrimination of nearly-identical distributions. Both our own tests and the literature indicate this piece is **method-independent and statistics-limited**: a foundation model (FM) will *not* out-classify a good high-level-feature NSBI on the κ_λ shape. The genuine leverage of an FM is elsewhere — **(i) data efficiency** at low signal stats, **(ii) signal-vs-background acceptance/purity**, and **(iii) robustness to systematics**. The recommended path is a *factorized, calibration-first* strategy: keep the validated signal-extraction machinery and do NSBI on high-level kinematics now; pursue a jet-free / event-level FM on the acceptance axis as a staged, calibration-gated upgrade.

1 Physics goal and current status

We aim to constrain the Higgs trilinear self-coupling modifier $\kappa_\lambda = \lambda_{HHH}/\lambda_{HHH}^{\mathrm{SM}}$ through non-resonant di-Higgs production in the $HH \to b\bar{b}\tau\tau$ final state, with the dominant sensitivity in the **resolved regime** ($m_{HH} \in [250, 400]$ GeV). κ_λ enters as a shape distortion of m_{HH} (interference between the triangle, $\propto \lambda_{HHH}$, and box amplitudes in ggF).

A neutral-simulation-based-inference (NSBI) fit using **10 event-level features**,

$$m_{HH},\ \cos\theta^\star,\ p_T^{HH},\ m_{bb},\ \Delta R_{bb},\ m_{\tau\tau},\ \Delta\phi_{HH},\ \Delta R_{\tau\tau},\ p_T^{H_1},\ p_T^{H_2},$$

already **tightens the κ_λ 95% CI by roughly a factor of two** (with $t\bar{t}$ background only, so far). The central *challenge* is statistical: very low signal yield, made worse once systematic uncertainties are profiled.

Hypothesis under test: a foundation-model + fine-tuning paradigm for the density-ratio (precision-classification) tasks inside the NSBI could improve the constraint — primarily via data efficiency and better object use.

2 The key decomposition: a κ_λ NSBI is two ML jobs

The single most clarifying point is that the measurement contains two distinct learning problems, which an FM helps very differently:
The κ_λ *constraint* comes from job (B). Job (A) sets the effective signal purity and is where the analysis is acceptance- and systematics-limited. A flavor-pretrained FM is built for (A), not (B) — which explains the results below.

Job	Discriminates	Information needed / best tool
(A) Signal vs. background	$HH \to b\bar{b}\tau\tau$ vs. $t\bar{t}$/QCD/fakes	*Object identity*: b-tagging, real-vs-fake τ, substructure — **flavor FM (Sophon-ak4) territory**.
(B) κ_λ shape / inference	$\kappa_\lambda = 0$ vs. 5 vs. … (morphing)	*Event kinematics* (m_{HH} shape) — high-level features / NSBI; a "precision classification of nearly-identical distributions" problem.

3 What we tested, and what we found

We adapted the `nsbi-lhc-toolkit` with a vendored Particle Transformer (ParT) backbone and **Sophon-ak4** (the AK4/resolved-jet ParT foundation model, pretrained for jet-flavor tagging on JetClass-II). A hierarchical model encodes each jet's constituents and aggregates jets+objects with an event set-transformer. Three controls were compared on the κ_λ=0 **vs.** κ_λ=5 task (with κ_λ=5 as the fixed reference/denominator): `scratch` (random-init ParT), `sophon_frozen`, `sophon_finetune`.

Table 1: *
Validation weighted-BCE (lower = better; random floor $= \ln 2 = 0.693$).

N / point	`scratch`	`sophon_finetune`	`sophon_frozen`
100k (40 ep)†	0.6378	**0.6351**	0.6486
2,000	**0.6752**	0.6775	0.6798
5,000	**0.6387**	0.6457	0.6538

†earlier configuration (before early-stopping and the lowered fine-tune LR); single seed at low N (error bars pending).

Empirical finding

On the κ_λ *shape* task, pretraining does **not** help — the ordering is consistently `scratch` $\leq$ `finetune` $<$ `frozen`. Frozen flavor features are the worst; fine-tuning merely relearns toward scratch.

Interpretation. (i) *Pretext mismatch*: Sophon-ak4 is optimized for jet flavor, whose per-jet embeddings are lossy about the p_T/mass needed to reconstruct m_{HH} from cross-jet correlations. (ii) The task is precisely "precision classification of nearly-identical distributions," which (next section) is statistics-limited and largely architecture-independent.

4 External evidence and the FM landscape

A 2025 review [1] organizes HEP foundation models into *supervised* (OmniLearn, Sophon, RS3L) and *self-supervised* (OmniJet-α, MPM) families. Points most relevant here:

- **Explicit vs. implicit priors are a wash — and statistics-limited** [2]. A direct comparison of L-GATr (Lorentz-equivariant) vs. OmniLearn (FM pretrained on 100M events) on *precision classification of similar classes* finds "comparable performance … given the statistical precision of the datasets," and that the gains from encoding physics are "method-independent." This benchmark *is* our κ_λ-shape task, and predicts our null result. Nuances:

on the H1/*ep* task the FM *beat* the equivariant net, and for small signal injections the implicit (FM) approach was slightly better — a mild lean toward FMs in the low-statistics corner.

- **Taus are not a blocker.** OmniJet-α fine-tuned for hadronic-τ reconstruction improved momentum resolution by $\sim$50% vs. from-scratch training [3] — so an earlier worry that OmniLearn-style FMs are unsuited to τ final states was premature.
- **Event-level FMs exist.** PECM [4] is a GNN over event objects (jets, e, μ, γ, MET with four-momenta + type IDs) pretrained on 120M events; it reports >5 percentage-point accuracy gains at 10^3–10^5 events/class (our regime) and $\sim$12× faster fine-tuning, transferring across simulators. Closest match to our event-level inputs.
- **Self-supervision retains kinematics; RS3L targets systematics.** MPM [5] learns task-agnostic representations (less biased than supervised flavor labels). RS3L [6] pretrains on *re-simulation* augmentations (varied shower/hadronization/detector), yielding embeddings *robust to those systematics* — directly relevant to a systematics-limited measurement.

5 Two architectural strategies for the measurement

Strategy A — jet-free, end-to-end. Feed *all* particle-flow candidates of the event (no jet clustering) into a ParT/event-FM, add the 10 high-level features, and do NSBI for κ_λ in one model. Motivated by a recent *jet-free* HH$\to 4b$ analysis [7]: ingesting all PF candidates ($\sim$300/event) into a ParT gave >5× search significance and an order-of-magnitude more retained signal at high background rejection. Caveats: it is *not* an FM (task-specific, trained on $\sim 10^8$ events — "training statistics play a critical role"), it is HH$\to 4b$ (a worse case for jet methods than $b\bar{b}\tau\tau$), and it required a *new* "calibrate–validate–measure" scheme (calibrate on $ZZ \to 4b$, validate on $ZH \to 4b$) because object-level scale factors do not apply to a jet-free discriminant.

Strategy B — factorized. Use the current CMS-style signal extractor for job (A), then NSBI on the high-level features (or an FM embedding) for job (B). The CMS $HH \to b\bar{b}\tau\tau$ analysis [8] already uses a sophisticated DNN: a Lorentz Boost Network front-end, a $\tau\tau$ neutrino-regression subnet, and a DenseNet classifier, with inputs that are full-system four-momenta *plus* ParticleNet/U-ParT (b-tag, c-vs-b, c-vs-l) and HHbTag scores — i.e. it already uses ParT as fixed tagger *scores* on reconstructed objects, with the entire validated systematics machinery built around those objects.

	A: jet-free FM (+10 feats) $\to$ NSBI	**B: CMS S/B $\to$ NSBI on feats/FM**
Information ceiling	Highest (soft radiation, no clustering/WP losses)	High, capped by reconstruction
Where it wins	**Signal acceptance & purity**	Clean κ_λ-kinematics (the proven $\sim$2×)
κ_λ-shape info	Same (kinematic)	Same (kinematic)
Calibration / systematics	**Hard & novel** (new scheme)	**Validated CMS machinery**
NSBI tractability	High-dim, harder closure/morphing	Low-dim, interpretable, easy closure
Data efficiency	Needs huge stats unless pretrained	Already efficient
Risk / time-to-result	High / frontier	Low / near-term

Important caveat: "Sophon-ak4 jet-free" mixes scales — Sophon-ak4 was pretrained on *single AK4 jets* ($\sim$128 constituents), so applying it to the *whole event* ($\sim$300 particles) is out-of-distribution. Jet-free-whole-event calls for an *event-scale* FM (PECM-style, or a from-scratch / SSL event-ParT), not Sophon-ak4 stretched beyond its domain.

6 Synthesis and recommendation

Key insights

1. The κ_λ constraint is **kinematic and statistics-limited**; explicit (equivariant) and implicit (FM) priors give comparable shape performance. Do not expect an FM to beat the high-level-feature NSBI on the κ_λ shape.
2. The FM's real value is **data efficiency**, **S/B acceptance/purity**, and **systematics robustness** — not shape extraction.
3. For a systematics-limited result, **calibration of any learned discriminant is the dominant risk**; the jet-free upside can be eaten by a poorly understood systematic.
4. Use the right FM for the right job: Sophon-ak4 (jet-scale) $\rightarrow$ per-jet S/B; event-scale FM $\rightarrow$ jet-free acceptance; high-level features / NSBI $\rightarrow$ κ_λ.

Recommended staged plan.

- **Near-term (low risk):** Strategy B, sharpened. Do NSBI on the high-level features (optionally + the CMS DNN score) within the validated signal regions — the direct productization of the $\sim 2\times$ result on real, calibrated data.
- **Test the right task:** a signal-vs-$t\bar{t}$ diagnostic (3 controls) — where per-jet Sophon *should* help, unlike the κ_λ-shape task. Plus a high-level kinematic ceiling baseline (BDT) to bound how much shape headroom exists.
- **Parallel R&D (high ceiling):** a jet-free / *event-level* FM for acceptance, introduced as a *calibratable* added discriminant (not a replacement), folded into the NSBI only once its calibration is validated.
- **Distinctive contribution:** RS3L-style re-simulation pretraining for a *systematics-robust* embedding — aimed squarely at the systematics bottleneck.

7 Open questions

- Do explicit and implicit priors learn the *same* physics, and does combining them (a Lorentz-equivariant *foundation* model) help in our regime?
- How large is the jet-free acceptance gain for $b\bar{b}\tau\tau$ (cleaner than $4b$), and can it be calibrated with $Z \rightarrow \tau\tau / Z \rightarrow b\bar{b}$ control samples?
- Does any S/B or acceptance gain actually translate into a tighter κ_λ CI once systematics are profiled (the figure of merit is the constraint, not BCE/AUC)?

References

[1] *Foundation models for high-energy physics* (review), arXiv:2509.21434.

[2] T. Wamorkar *et al.*, *Explicit or Implicit? Encoding Physics at the Precision Frontier*, arXiv:2603.08802.

[3] *Reconstructing hadronically decaying tau leptons with a jet foundation model* (OmniJet-α), arXiv:2503.19165, SciPost Phys. Core (2025).

[4] *Pretrained Event Classification Model for High Energy Physics Analysis*, arXiv:2412.10665.

[5] *Masked Particle Modeling on Sets: Towards Self-Supervised HEP Foundation Models*, arXiv:2401.13537.

[6] *Re-Simulation-based Self-Supervised Learning for Pre-Training Foundation Models*, arXiv:2403.07066; Phys. Rev. D **111**, 032010 (2025).

[7] *A calibratable, jet-free framework for* $HH \to 4b$ *via all-particle multiclass classification*, arXiv:2508.15048.

[8] CMS Collaboration, *Search for non-resonant Higgs boson pair production in the* $b\bar{b}\tau\tau$ *final state with Run-3 data*, CMS AN-25-103 (draft, 2026).

[9] O. Amram, K. Mikuni, B. Nachman *et al.*, *OmniLearn*; *OmniJet-*α, arXiv:2403.05618.

[10] *A Lorentz-Equivariant Transformer for All of the LHC* (L-GATr), arXiv:2411.00446; *Lorentz-Equivariant Geometric Algebra Transformers*, arXiv:2405.14806.